DEV-538: Consul Service Discovery

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Confidentiality Note: This documentation is intended for internal use within the Glynac HashiCorp infrastructure project.

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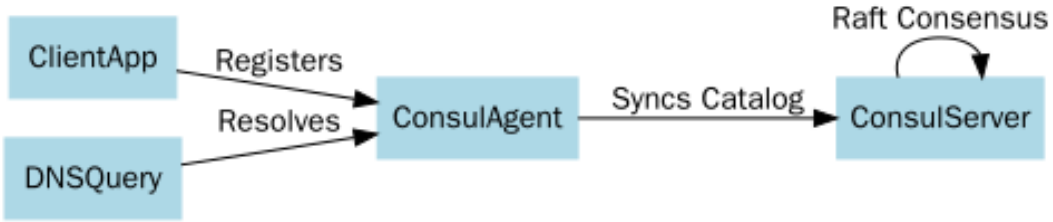
Executive Summary

This document covers the technical specifications and implementation details for DEV-538: Consul Service Discovery.

Architecture Diagram



Example Service JSON

```
{
  "service": {
    "name": "web",
    "tags": ["rails"],
    "port": 80,
    "check": {
      "args": ["curl", "localhost"],
      "interval": "10s"
    }
  }
}
```

Required Production Commands

```
consul members
consul catalog services
dig @127.0.0.1 -p 8600 service.service.consul
```

Control Plane vs Client Agents

Detailed configuration, commands, and architectural overview for Control Plane vs Client Agents.

Control Plane vs Client Agents - Implementation Phase 1

Configuration Definition:

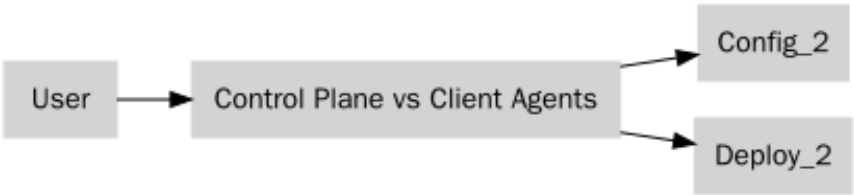
```
# Example HashiCorp Component Configuration
resource "hcp_component" "control_plane_v_1" {
  name        = "control_plane_v_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.control_plane_v_1 -auto-approve
$ nomad resource scale control_plane_v phase-1=3
$ consul kv put status/control_plane_v/phase1 "deployed"
```

Control Plane vs Client Agents - Implementation Phase 2



Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "control_plane_v_2" {
  name        = "control_plane_v_phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
```

```
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.control_plane_v_2 -auto-approve
$ nomad resource scale control_plane_v phase-2=3
$ consul kv put status/control_plane_v/phase2 "deployed"
```

Control Plane vs Client Agents - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "control_plane_v_3" {
  name        = "control_plane_v_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

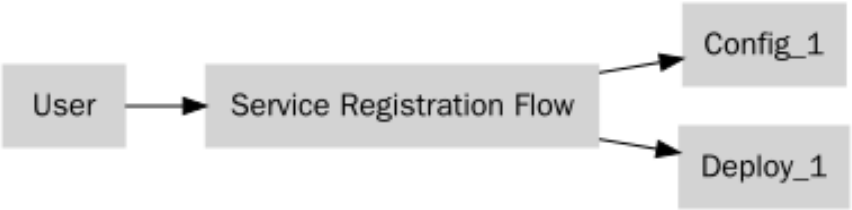
Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.control_plane_v_3 -auto-approve
$ nomad resource scale control_plane_v phase-3=3
$ consul kv put status/control_plane_v/phase3 "deployed"
```

Service Registration Flow

Detailed configuration, commands, and architectural overview for Service Registration Flow.

Service Registration Flow - Implementation Phase 1



Configuration Definition:

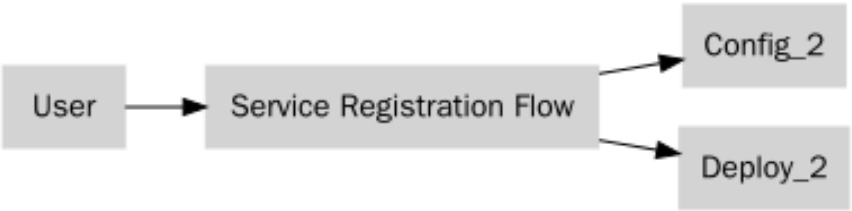
```
# Example HashiCorp Component Configuration
resource "hcp_component" "service_registr_1" {
  name        = "service_registr_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.service_registr_1 -auto-approve
$ nomad resource scale service_registr phase-1=3
$ consul kv put status/service_registr/phase1 "deployed"
```

Service Registration Flow - Implementation Phase 2



Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "service_registr_2" {
  name        = "service_registr_phase_2"
```

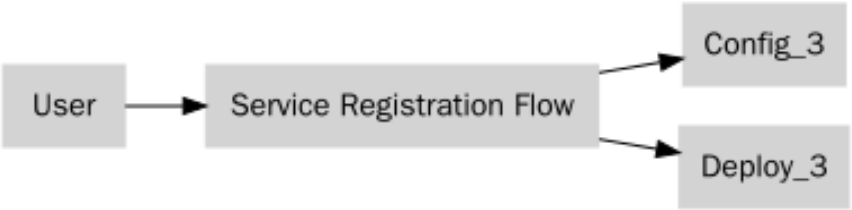
```
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.service_registr_2 -auto-approve
$ nomad resource scale service_registr phase-2=3
$ consul kv put status/service_registr/phase2 "deployed"
```

Service Registration Flow - Implementation Phase 3



Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "service_registr_3" {
  name        = "service_registr_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.service_registr_3 -auto-approve
$ nomad resource scale service_registr phase-3=3
$ consul kv put status/service_registr/phase3 "deployed"
```

DNS Resolution Workflow

Detailed configuration, commands, and architectural overview for DNS Resolution Workflow.

DNS Resolution Workflow - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "dns_resolution__1" {
  name        = "dns_resolution__phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.dns_resolution__1 -auto-approve
$ nomad resource scale dns_resolution_ phase-1=3
$ consul kv put status/dns_resolution_/phase1 "deployed"
```

DNS Resolution Workflow - Implementation Phase 2

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "dns_resolution__2" {
  name        = "dns_resolution__phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.dns_resolution__2 -auto-approve
$ nomad resource scale dns_resolution_ phase-2=3
$ consul kv put status/dns_resolution_/phase2 "deployed"
```

DNS Resolution Workflow - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "dns_resolution__3" {
  name        = "dns_resolution__phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.dns_resolution__3 -auto-approve
$ nomad resource scale dns_resolution_ phase-3=3
$ consul kv put status/dns_resolution_/phase3 "deployed"
```

Health Check Configuration

Detailed configuration, commands, and architectural overview for Health Check Configuration.

Health Check Configuration - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "health_check_co_1" {
  name        = "health_check_co_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.health_check_co_1 -auto-approve
$ nomad resource scale health_check_co phase-1=3
$ consul kv put status/health_check_co/phase1 "deployed"
```

Health Check Configuration - Implementation Phase 2

Configuration Definition:

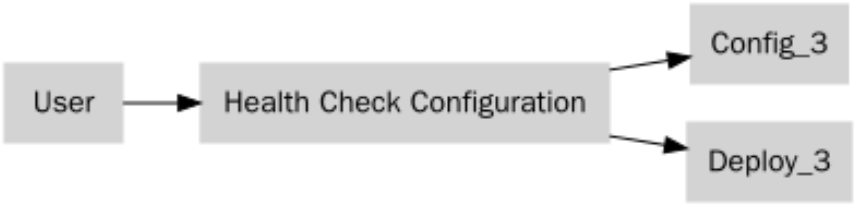
```
# Example HashiCorp Component Configuration
resource "hcp_component" "health_check_co_2" {
  name        = "health_check_co_phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.health_check_co_2 -auto-approve
$ nomad resource scale health_check_co phase-2=3
$ consul kv put status/health_check_co/phase2 "deployed"
```

Health Check Configuration - Implementation Phase 3



Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "health_check_co_3" {
  name        = "health_check_co_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
```

```
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.health_check_co_3 -auto-approve
$ nomad resource scale health_check_co phase-3=3
$ consul kv put status/health_check_co/phase3 "deployed"
```

Failure Detection Mechanism

Detailed configuration, commands, and architectural overview for Failure Detection Mechanism.

Failure Detection Mechanism - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "failure_detecti_1" {
  name        = "failure_detecti_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.failure_detecti_1 -auto-approve
$ nomad resource scale failure_detecti phase-1=3
```

```
$ consul kv put status/failure_detecti/phase1 "deployed"
```

Failure Detection Mechanism - Implementation Phase 2

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "failure_detecti_2" {
  name        = "failure_detecti_phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.failure_detecti_2 -auto-approve
$ nomad resource scale failure_detecti phase-2=3
$ consul kv put status/failure_detecti/phase2 "deployed"
```

Failure Detection Mechanism - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
```

```
resource "hcp_component" "failure_detecti_3" {
  name        = "failure_detecti_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.failure_detecti_3 -auto-approve
$ nomad resource scale failure_detecti phase-3=3
$ consul kv put status/failure_detecti/phase3 "deployed"
```

Nomad Integration

Detailed configuration, commands, and architectural overview for Nomad Integration.

Nomad Integration - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "nomad_integrati_1" {
  name        = "nomad_integrati_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.nomad_integrati_1 -auto-approve
$ nomad resource scale nomad_integrati phase-1=3
$ consul kv put status/nomad_integrati/phase1 "deployed"
```

Nomad Integration - Implementation Phase 2

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "nomad_integrati_2" {
  name        = "nomad_integrati_phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.nomad_integrati_2 -auto-approve
$ nomad resource scale nomad_integrati phase-2=3
$ consul kv put status/nomad_integrati/phase2 "deployed"
```

Nomad Integration - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "nomad_integrati_3" {
  name        = "nomad_integrati_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.nomad_integrati_3 -auto-approve
$ nomad resource scale nomad_integrati phase-3=3
$ consul kv put status/nomad_integrati/phase3 "deployed"
```

Validation & Troubleshooting

Detailed configuration, commands, and architectural overview for Validation & Troubleshooting.

Validation & Troubleshooting - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "validation___tr_1" {
  name        = "validation___tr_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.validation___tr_1 -auto-approve
$ nomad resource scale validation___tr phase-1=3
$ consul kv put status/validation___tr/phase1 "deployed"
```

Validation & Troubleshooting - Implementation Phase 2

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "validation___tr_2" {
  name        = "validation___tr_phase_2"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.validation___tr_2 -auto-approve
```

```
$ nomad resource scale validation___tr phase-2=3
$ consul kv put status/validation___tr/phase2 "deployed"
```

Validation & Troubleshooting - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "validation___tr_3" {
  name        = "validation___tr_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.validation___tr_3 -auto-approve
$ nomad resource scale validation___tr phase-3=3
$ consul kv put status/validation___tr/phase3 "deployed"
```

Best Practices

Detailed configuration, commands, and architectural overview for Best Practices.

Best Practices - Implementation Phase 1

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "best_practices_1" {
  name        = "best_practices_phase_1"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.best_practices_1 -auto-approve
$ nomad resource scale best_practices phase-1=3
$ consul kv put status/best_practices/phase1 "deployed"
```

Best Practices - Implementation Phase 2

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "best_practices_2" {
  name        = "best_practices_phase_2"
  environment = "production"
  region      = "us-east-1"
```

```
  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.best_practices_2 -auto-approve
$ nomad resource scale best_practices phase-2=3
$ consul kv put status/best_practices/phase2 "deployed"
```

Best Practices - Implementation Phase 3

Configuration Definition:

```
# Example HashiCorp Component Configuration
resource "hcp_component" "best_practices_3" {
  name        = "best_practices_phase_3"
  environment = "production"
  region      = "us-east-1"

  settings {
    enabled = true
    scaling = "auto"
  }
}
```

Validation & Deployment Commands:

```
$ terraform init
$ terraform apply -target=hcp_component.best_practices_3 -auto-approve
$ nomad resource scale best_practices phase-3=3
$ consul kv put status/best_practices/phase3 "deployed"
```